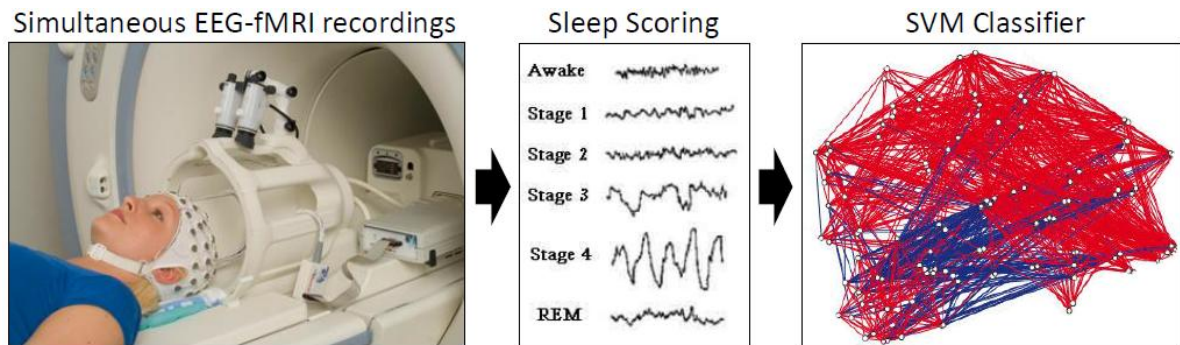
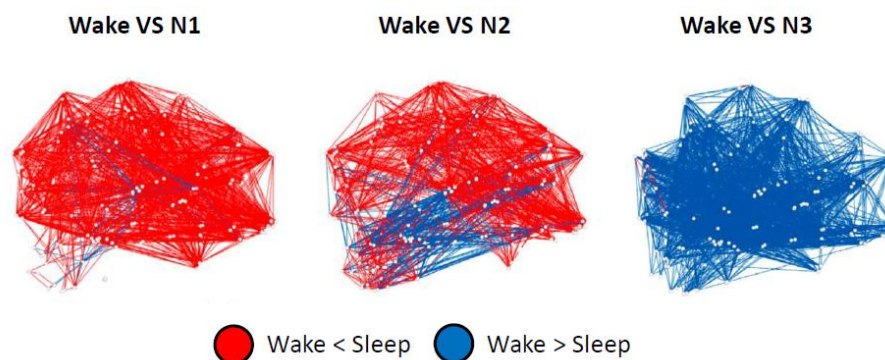


The advantage of combining different techniques is that you can take the advantage of each of them. EEG with MRI is pretty basic, you just need an EEG that is MRI compatible (not by definition), especially the amplifier is not MRI compatible. Most importantly this combination allows to monitor the vigilance of the subject, how sleepy they are, because right now the only method we have to measure sleep is through EEG. The machine learning classifier is able to recognize the EEG stages based on the f-MRI connectivity.



What they found is that if you do an MRI scan of 10 minutes, 50% of the subjects will be asleep at the end.



The net reduction in connectivity observed during deep sleep reflect a reduction in brain interregional cross talk, i.e., a loss of global integration well in line with a phenotypically concomitant reduction in consciousness. Modularity, a spatial measure of functional segregation, generally increased across the brain with deepening sleep.

The issue if the subject fall asleep is that first of all data is not constant over-time, not uniform. But the biggest problem is that some fall asleep and others not, you can't compare them. People with social anxiety difficultly fall asleep knowing that someone is watching them. In some systems you can have eye-tracking, so that you can always know if the patient is still awake.

Vigilance and awareness

Sleep is important for memory processing, metabolism, tissue repair, memory processing, immune system, mood. When you sleep after something happen two things happen, the memory will be stabilized in the brain (increase of the signal to noise ratio), and sleep operates on mood, emotional regulation. When you sleep on it the reason that maybe angered you before sleeping it seems now softer, thanks to sleep.

When you have a patient that is in a vegetative state, they don't fall in a state of sleeping, it's when you see that they finally fall asleep that there can be possibilities for them to wake up and recover.

The brain has a sort of special system that is different from any other part of the body, the glymphatic system. In the brain there is no lymphatic system, but researchers saw that during sleep something was changing, the shape and position of the astrocytes, creating space from one cell to another, canals used to move around fluids that remove from the brain metabolized things, if you don't remove them the brain saturate. This system doesn't work properly in patients who develop Alzheimer disease.

In an experiment some people had to do some tasks, after sleeping their performance improved, with respect to the deepness of their sleep. Deeper sleep has more slow waves, and these slow waves were specifically located in the regions of the brain involved in the learning process.

With another experiment it was found that if the subject practice and then slept, he was much more likely to find the solution to the problem compared to the people that didn't.

Experiment: Randy Gardner

Randy Gardner (born 1947) is the holder of the scientifically documented record for the longest a human has intentionally gone without sleep not using stimulants of any kind. In 1964, Gardner (at the time high school student in San Diego, California) stayed awake for 264.4 hours (11 days 24 minutes). The SD led to serious cognitive and behavioural changes:

- Alterations of mood.
- Problems with concentration.
- Problems with short term memory.
- Paranoia.
- Hallucinations.

On the eleventh day, when he was asked to subtract seven repeatedly, starting with 100, he stopped at 65. When asked why he had stopped, he replied that he had forgotten what he was doing. It was seen also that the need of sleep put us in a negative mood.

For children the effect of not sleeping is worst, because they are in a period of development in which the brain really needs sleep. You can die after months without sleeping.

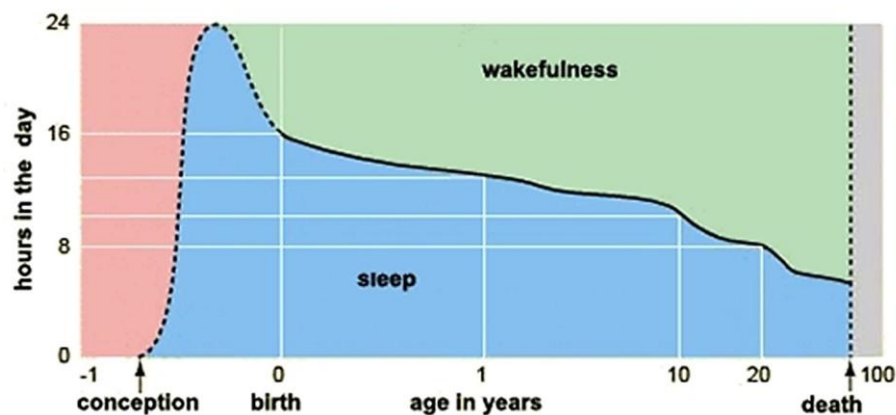
Fatal familial insomnia (FFI) is an extremely rare autosomal dominant inherited prion disease of the brain. It is usually caused by a mutation to the protein PrPC but can also develop spontaneously in patients with a non-inherited mutation variant called sporadic fatal insomnia (sFI). FFI has no known cure and involves progressively worsening insomnia, which leads to hallucinations, delirium, confusional states, and eventually, death. The average survival time for patients diagnosed with FFI after the onset of symptoms is 18 months.

We can live with also few hours of sleep, but there are going to be alterations in behaviour. Performances are affected by the need of sleep.

For older people the amount of sleep they are able to maintain decreases, but precisely because the brain can work with less sleep, but only because you are able to maintain sleep for lesser time.

Humans sleep about 8 hours a day, but it depends on the age. New births sleep also for 16 hours, very often non continuously, but then at some point sleep stabilizes (1-2 years of age) decreasing

and the reaching 8 hours around adolescence. When we are younger (from 10 to 20 years old) we need to sleep more, because our brain needs it for memory and plasticity to occur. In fact our brain becomes lesser plastic, the capability to learn and adapt to new environments decreases over-time.



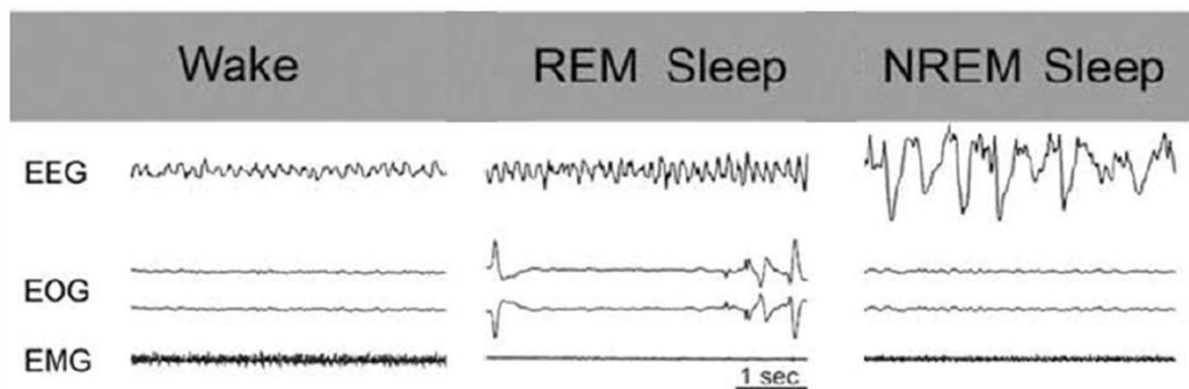
Sleeping too much (more than 10-12 hours per day) is also bad, if it happens multiple times. If you do that you have higher probability of developing some metabolic or other diseases.

Definition of sleep

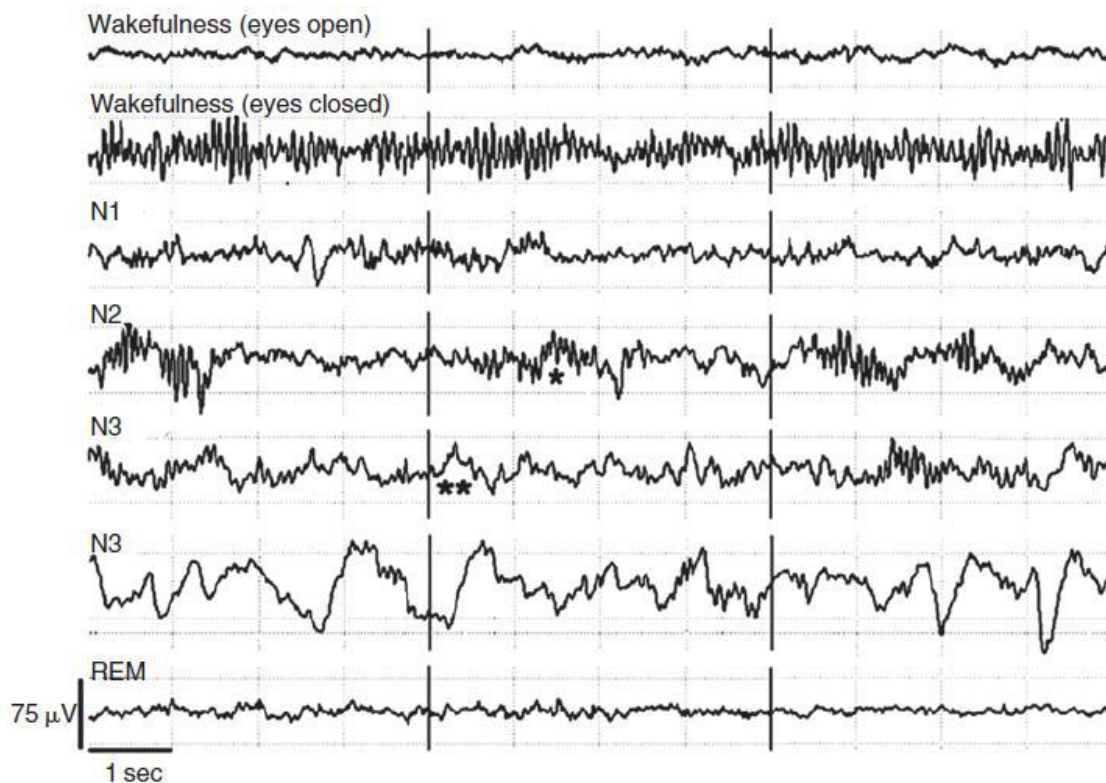
Behavioural: Behavioural state with characteristic immobile posture and diminished but readily reversible sensitivity to external stimuli.

But this definition is not so efficient, because the person may be in a vegetative or come state, not sleeping at all or just faking. So, we need a marker to define whether someone is sleeping or not and this comes with EEG. We can use also other measures, like temperature of the skin or heart rate, but the EEG is the most reliable.

Electrophysiological:



We typically use three signals, because we need to distinguish different conditions. The wake EEG comprehends very fast activity; the eyes movement are voluntarily controlled and there is high muscular tone. Then when I enter the REM sleep, the activity may be similar in terms of EEG, but in the EOG I get a non-voluntarily controlled eye movement, and the muscular tone is decreased. That is why I need these two additional signals. Non-REM sleep is recognizable because there is no eye movements, the muscular tone is intermediate, and the brain goes on-off all the time, completely different pattern of activity.



In the first line we can see that wakefulness with the eyes open is dominated by a fast activity, but then when I close my eyes α -activity appears, this is called the α -block and it happens because this activity is mainly produced by the occipital cortex, by the visual regions in the brain, and is produced when the visual regions are not receiving inputs. If the subject is tired, sleepy, his α -activity will be larger when he closes his eye and will also spread to more anterior regions (frontal).

Then at some point if I try to fall asleep this α -activity will disappear and the pattern of activity will slow down, going to another stage called non-Rem state (N1, N2, N3, N3 (N4 at first)). When you go from N1 to N3 basically the brain slows down, the oscillation is slower in frequency and bigger in amplitude. Then when you have REM sleep you go back to a faster activity.

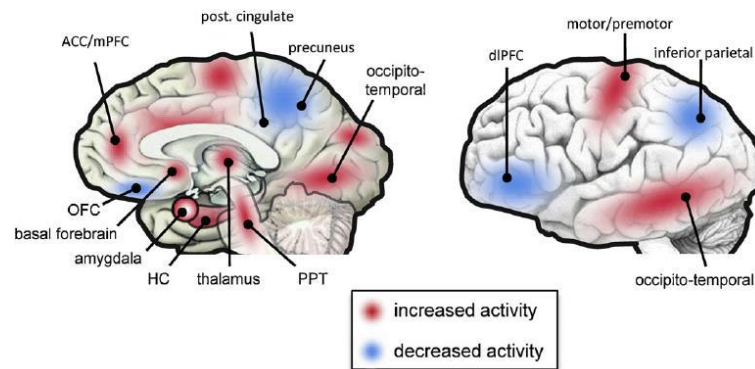
REM sleep is not a state that you reach immediately, typically happens after 90 minutes when you fall asleep, specially at night.

It seems that NREM is more important than REM from a functional perspective, right now the researchers are trying to amplify the number of slower waves, so that I can sleep less than 7 hours. Drugs and anaesthesia drugs don't work. Right now, sensory stimulation is being used, the reason is that basically when you are in sleep already, the brain is trying to maintain sleep, so if you give a stimulus not too strong, the brain will generate more big slow waves. Trying it the professor didn't find an advantage from it.

Changes in brain activity

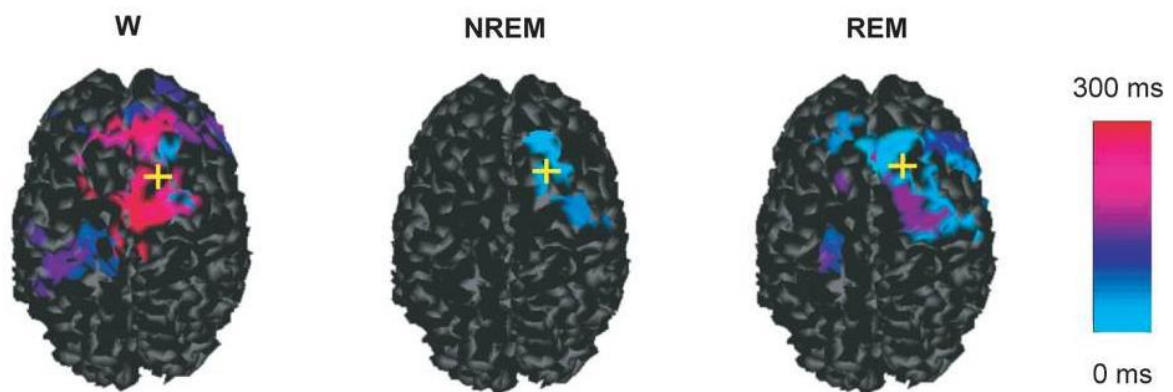
There are different arousal systems activated in the brain that all turn off when you fall asleep, with one exception, acetylcholine. During sleep there is a reduction in metabolism in the brain, but not generalized, in NREM sleep there is a reduction in metabolism in regions that overlap in

part the default mode network and in part the thalamus. Sleep is not a state of overall reduction of metabolism.



Overall, REM sleep seems to be in a more active state, there are a lot of things going on. In REM sleep there is a lot of activity in the limbic system, in the hippocampus and the amygdala, regions that process memories and emotions, sleep is important for memory consolidation and emotional processing. There are theories that say that REM sleep at least decreases the emotional components in memories.

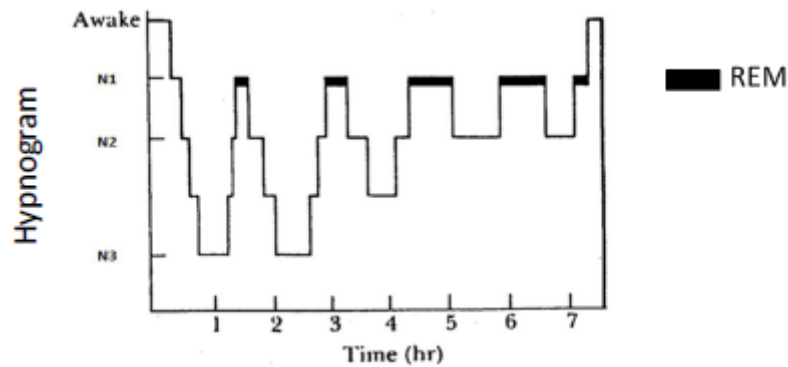
When we look at the states, we see that they have different characteristics in terms of complexity.



In this study they used TMS+EEG. The yellow cross is the site where they applied transcranial magnetic stimulation and the colour represents the response of the brain to the stimulation; red is late response. When the subject is awake the stimulus flows across the different regions forth and back in different directions, but if you do the same in NREM sleep you have a short lasting and very simple response, in which most of the brain response remains localized. And finally in REM sleep you have something that is in between, you get a more complex response.

Sleep cycle

The hypnogram represents the distribution of the different sleep stages during the night. On the x-axis you have the time since falling asleep and in the y-axis the time spent in the different stages. You start awake and then into N1, N2, N3, you spend some time into N3 sleep and then you go back to REM sleep, and this is repeated several times. Each time, the amount of time spent in N3 decreases and is especially strong at the beginning of the night. Young people are much more efficient than older ones, so if you take someone around 20-25 years old they will produce a lot of huge slow waves, so a lot of N3 sleep, but when you get older the amount of N3 sleep decreases.



The succession of NREM sleep stages followed by an episode of REM sleep is called a sleep cycle and lasts approximately 90–110 minutes in humans. There is a total of 4–5 cycles per night. Slow wave sleep (N3) is prominent early in the night, especially during the first sleep cycle, and diminishes as the night progresses. As slow wave sleep wanes, periods of REM sleep lengthen and show greater phasic activity. A healthy young adult typically spends about 5% of the sleep period in stage N1, about 50% in stage N2, 20–25% in stage N3, and 20–25% in REM sleep.

The first two cycles are especially important, because they contain most of the NREM N3 sleep, and this is why if you really need to sleep is better to sleep at least 2-3 cycles.

Sleep regulation

The less I sleep the more my brain accumulates sleep need. The homeostatic factor of sleep regulation just tells you that you need to sleep, and the amount of sleep you need depends on how much you have slept before.

The other concept that is crucial is the concept of circadian regulation, and these two regulatory factors interact with each other, to determine your sleep propensity during the day. Û

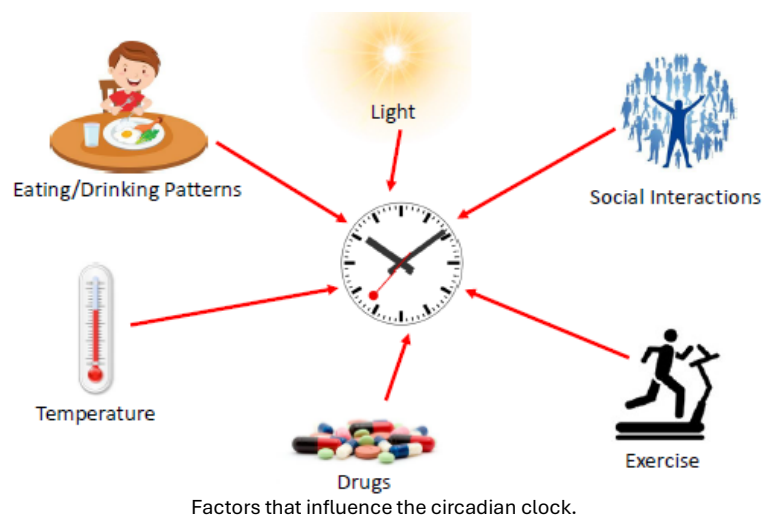
In sample terms circadian regulation is our internal clock. We have cells and systems that regulate the work of our body with an autonomy of 24 hours, but others can be also tuned in a slightly different rhythm, maybe of 24.5/25 hours. But then these genes and system can be entrained, being regulated by exposure to light.

In our retina there are specific receptors that are different from the receptors to see, and these are especially sensitive to blue light, and they send a signal to a nucleus (suprachiasmatic nucleus), located in the anterior hypothalamus. This nucleus regulates the release of a hormone called melatonin that is produced by the pineal gland. In normal conditions, after the beginning of the dark period you have an increase in release of melatonin, and its circulation signals the transition to sleep, determining drowsiness and a reduction in the core body temperature. Right now (this period with phone, tv etc.), the presence of light sources can affect and delay the release of melatonin, reducing the quality of our sleep.

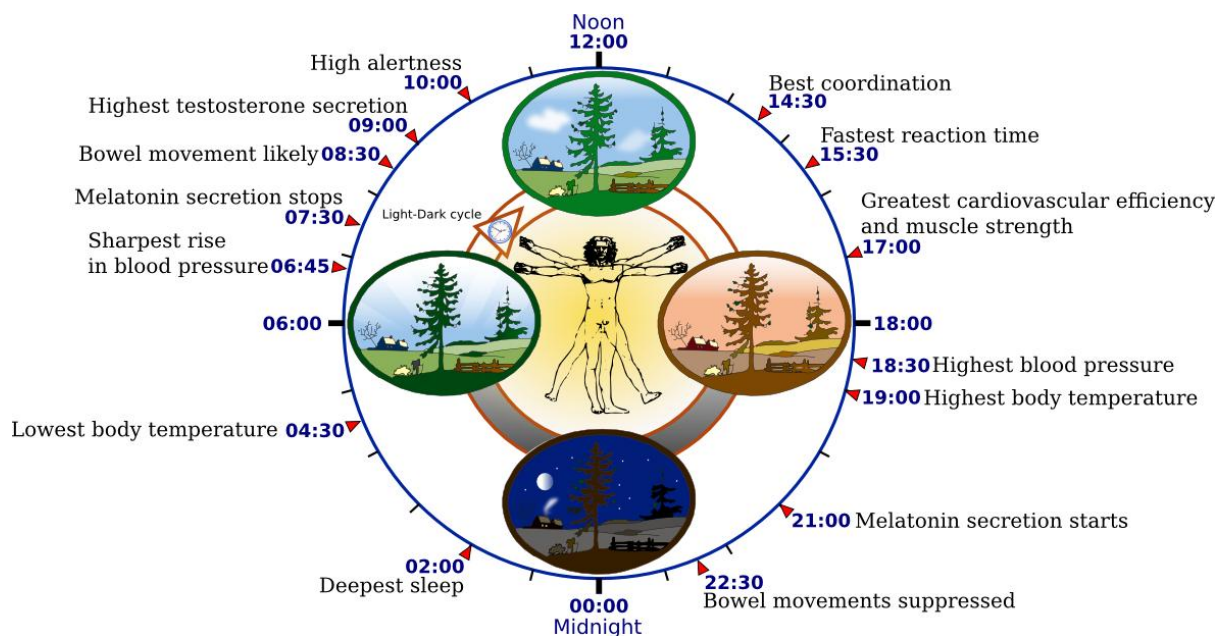
Exposure to light is a factor that regulates the release of melatonin, because our internal clock is not regulated exactly in 24 hours. Light is not the only factor, others contribute to align the circadian clock to the actual time, if we don't do this alignment our cycle of melatonin will follow the normal circadian clock (not normally aligned).

- The rhythm has an endogenous free-running period that lasts approximately 24 hours.
- The rhythms are entrainable (zeitgeber).

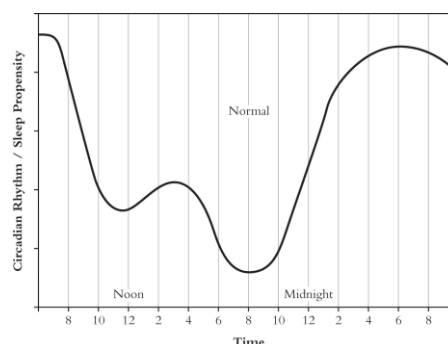
- The rhythms exhibit temperature compensation (maintained circadian periodicity).



All functions in the body are all affected by the circadian clock, they are not under our control.

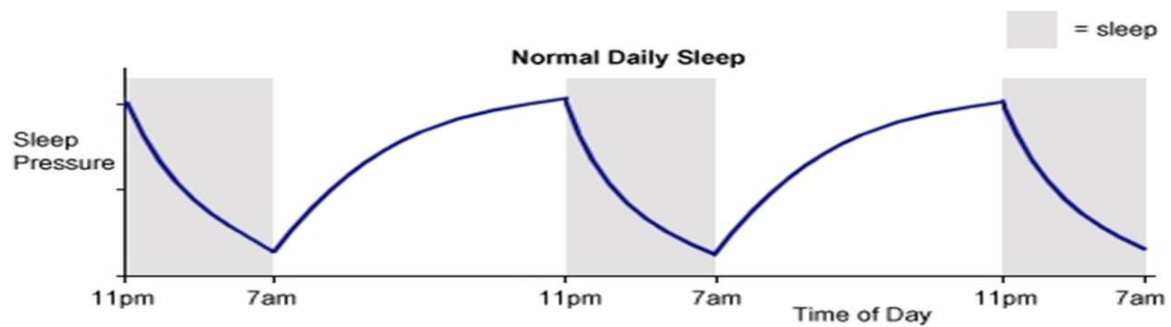


Sleep propensity is modulated by the circadian clock. Starting from midnight, is the moment in which sleep propensity is the lowest (non-sleep zone), and it then starts to increase reaching a peak around 3-5 in the morning, and then during the night it starts to decline again. During the day, after launch, sleep propensity starts again to increase, related also to the digestion process but is also something that is really inside our circadian system.

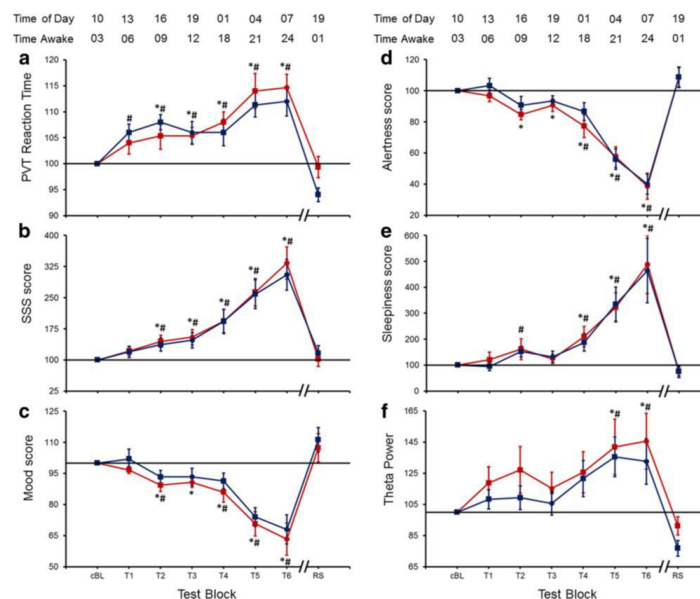


Homeostatic sleep regulation

The more we stay awake the more we accumulate sleep need. But what drives this homeostatic factor? We think it could be the accumulation of metabolized, that are produced by the brain activity. It has been seen that adenosine is associated with the increase of sleep need (produced by the energetic consumption), reaching specific receptors on cells and makes us sleepy. This is why coffee works, because it binds the same receptors and blocks these receptors, but you can't drink too many coffees, because adenosine will accumulate and then when you stop drinking coffee you will feel worst.



The homeostatic sleep need and the circadian regulation interact together.



In this example, in which the speed of reaction time was measured, you can see that it goes up and then down exactly in the time where the circadian propensity to sleep is at the lowest, so at 19 my performances improve, and then after the non-sleep zone passes my sleep need will increase again.

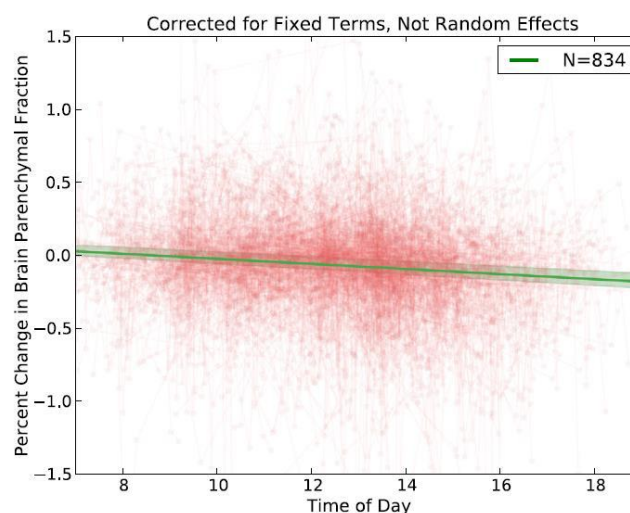
The same kind of pattern can be seen also for behaviour, mood. For the last one they asked people to rate their mood during the day (every 2 hours), and it tends to go down at the beginning of the task and then they recover after lunch, and finally it goes down again during sleep before recovering after sleeping. In this experiment they went sleeping in the morning, not the best because I am much less likely to sleep well. If I go to sleep outside the circadian optimal window the sleep will never be optimal.

Looking for the other graphs (alertness, sleepiness), they also follow the same pattern, they felt sleepy in the afternoon and then it decreases in the evening.

The last graph represents instead brain activity, from which we can see that it follows exactly the same pattern, in particular here we looked to low frequency activity, not exactly as low as sleep frequency, but slower than normal frequency. Delta activity (4-9 Hz).

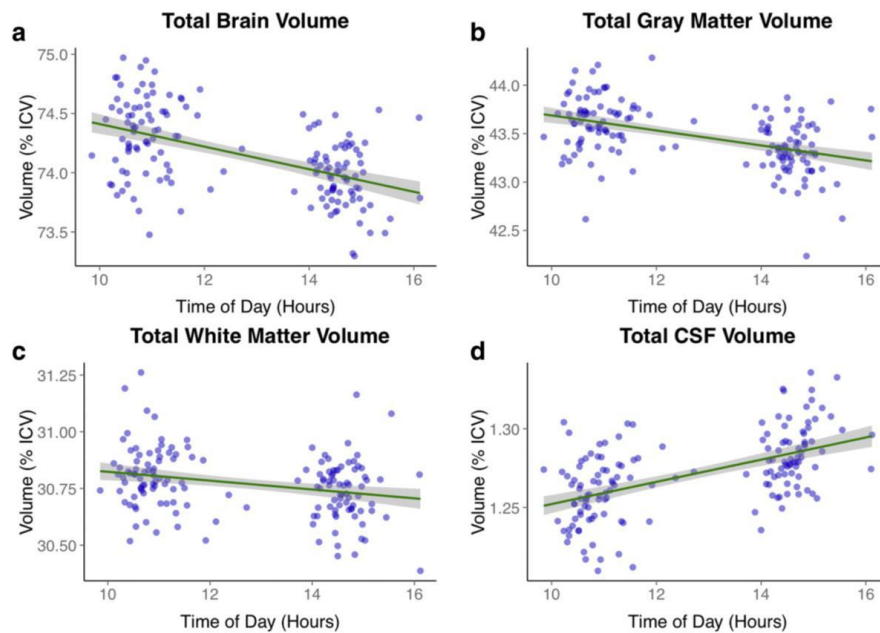
Brain activity changes during the day, if you measure it during the day, you will get different pattern of brain activity. Also apparently, in the brain, different regions have different sensitivity to the circadian regulation, showing a modulation of activity that is differently aligned with the circadian clock. Brain activity is not constant during the day.

A study published showed that the brain structure changes across the day, the measures of the brain structure that we have. What we get measuring is a magnetic resonance image, an image that reflect not only the characteristics of the brain tissue but also water distribution. The study took the data from a database, comprehending the time of the day, and looked at the brain size (volume), measured with the techniques that we normally use to analyse MRI data, across the day. What they found is that, if you correct for all possible variability across individuals, you seem to see a reduction in brain size across the day.



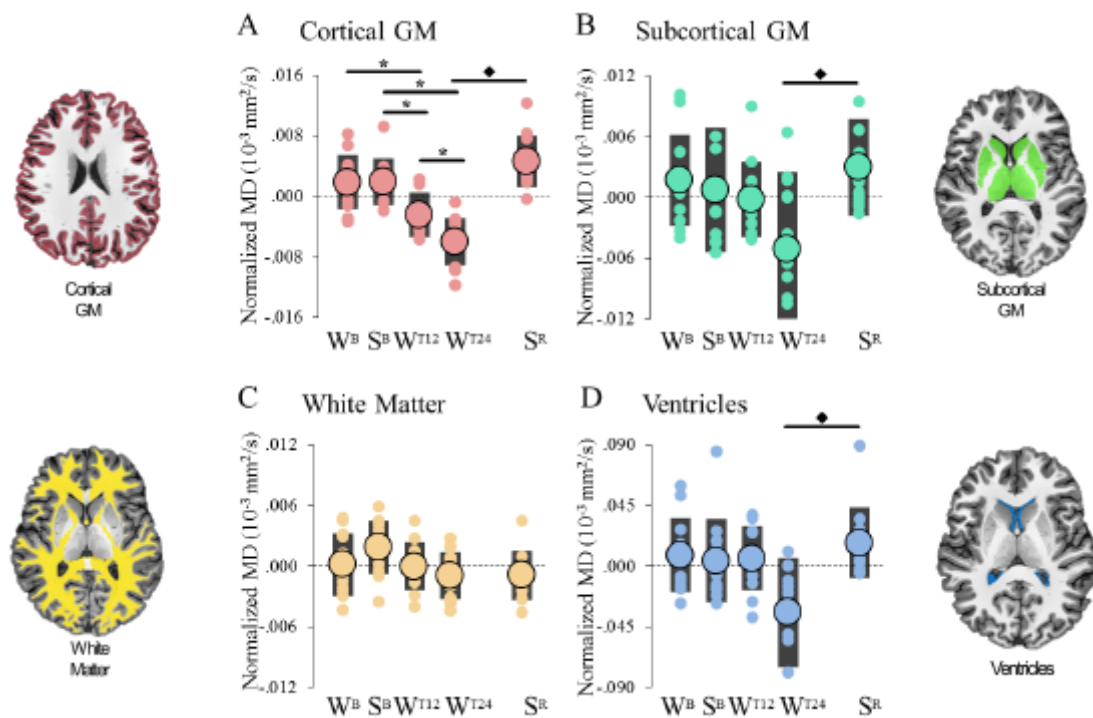
It was seen in another studies that if you practice with a task there is an increase in brain matter volume specifically in regions related to the task, so apparently these changes may not reflect just water distribution but real anatomical changes. So, we don't know actually, because we have only this indirect measure. If you do an experiment, it's important that data are extracted at the same time of the day.

In other studies, it was shown that even with only 4 hours of difference from one scan to another, it is associated a difference in brain volume that you can measure. There is only one thing that increased, total CSF volume, intracranial volume for CSF (amount of volume), so in this case the changes that we see are not really anatomical but related to a redistribution of water in the brain. Water basically moves from inside the brain to around the brain, when we go from the morning to the evening.



For another study of the professor, MRI scans were taken from Friday evening and then every 12 hours. What you can see is that when the subject stays in the lab doing task for 24 hours the brain shrinks and their cortical grey matter decreases, increasing again after sleep.

Measures of how much water can move in the tissue that you are analysing were also taken for different regions. In the first case the decrease means that in the cortical grey matter water was freer to move, if you increase the number of synapsis or number of cells water will have less freedom of moving because there is less space. If you learn to do the task, more space in the tissue will be occupied by synapsis for instance and so water will have less movement.



The key point is that brain activity and brain structure measures changes across the day.

Implications for neuroimaging

All subjects should be examined at around the same time of day (If possible, in a time window not larger than 1-2 hours). It is good practice to report in your article at what time of day the data was collected and whether consistency across subjects and groups was ensured!

It's also important to choose the best window of the day in which collect my data, and that is in the morning around 10-13, when the subject is fully awake and alert. The circadian pressure is low and also the homeostatic pressure is relatively low, so I can expect the performance to be maximal. It depends also on the habits of the participants; you have to match the circadian alignment of the person.

Overall vigilance and sleep quality may be assessed using simple questionnaires (useful to exclude extreme cases). Actigraphy recordings may be used to ensure no between-group differences in sleep time before the experiment, or to exclude a lack of circadian alignment.

Another instrument that is very useful (present also in smartphones) is the actigraphy. It is an accelerometer that looks at the activity of the arm and tells you if the subject is still, sleeping or moving. You can measure the circadian rhythm of the subject just looking when they wake up and fall asleep. It can also control the exposure to the light, temperature, heart variability and say the different stages of sleep.

First you have to indicate the beginning and the end of the window in which the subject was more likely to be sleeping and then the software calculates, based on the amount and amplitude of movement, the amount of sleep that the subject has. With this measures you can see how stable the time is the subject wakes up and fall asleep, and you can measure the central point in this window that is an indication of the circadian alignment of your participants.

If group-level differences cannot be excluded, the best practice is to monitor the state of vigilance during the MRI session and discard subjects or segments including behavioural sleep. Though it is impossible to avoid the noise.

It can be helpful to ask the patient to keep the eyes open, you can do an EEG-fMRI co-registration for sleep scoring, and you can do shorter acquisitions.

The difference between wakeness and sleep is not just the time we spend in bed but sleep is part of wakeness. Sleep is regulated across it, our vigilance change in the day as well as our sleep need, and there are specific mechanisms that regulate it and these have an effect on behaviour, brain activity and even brain structural measures.